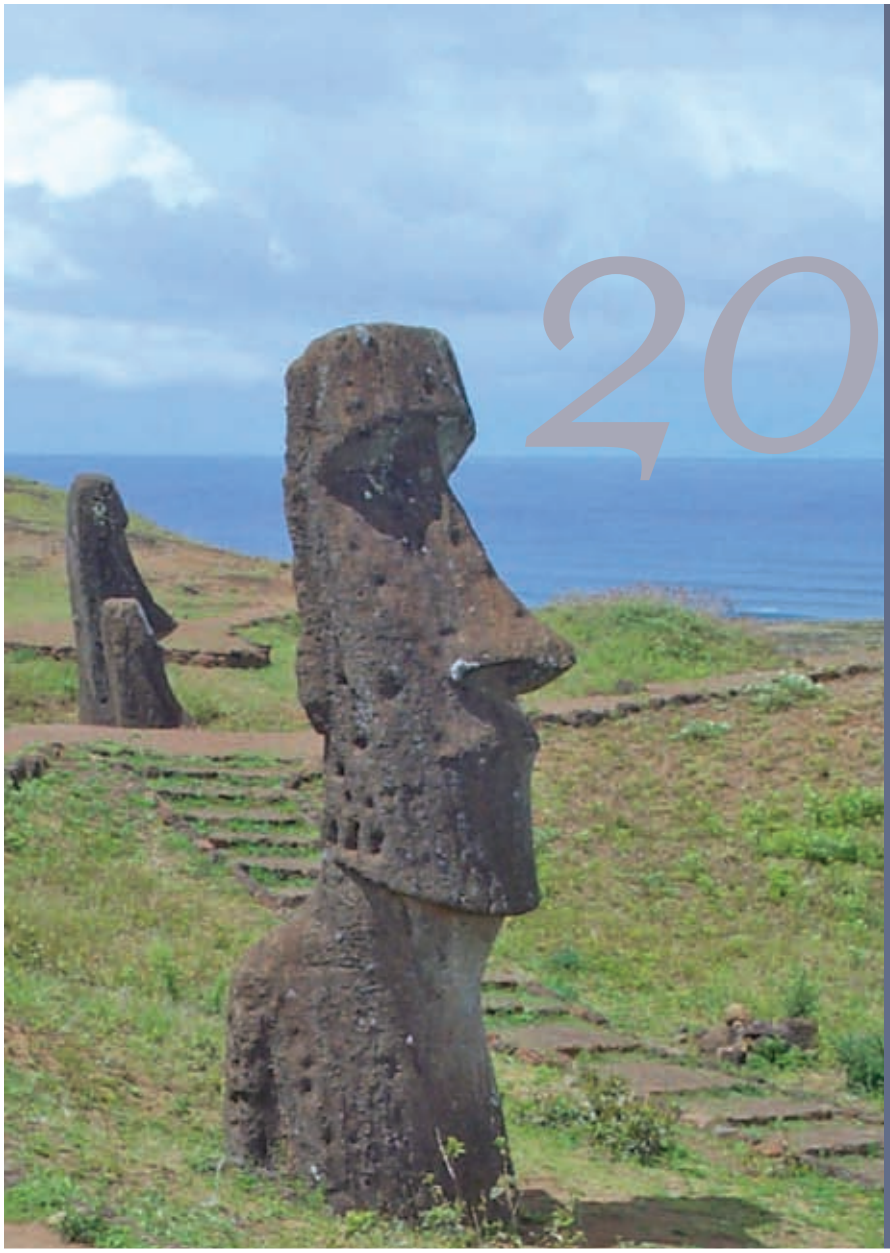

SOUTH AMERICA

2008

Easter Island Cultures

Dr. Christopher M. Stevenson
Sonia Haoa
Virginia Department of Historic Resources

Earthwatch.

2008 EXPEDITION BRIEFING

EARTHWATCH INSTITUTE
FINDING SOLUTIONS FOR A SUSTAINABLE FUTURE

Our Mission

Earthwatch Institute engages people worldwide in scientific field research and education to promote the understanding and action necessary for a sustainable environment.

We believe that achieving a sustainable future requires objective scientific data from the field—and that the scientific process must engage the general public if it is to change the world. To that end, we involve people from all walks of life directly in global field research.

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The Expedition Briefing in your hands is your initiation to the project from the scientist's own perspective. You will learn about the inspiration that led the scientist(s) to launch the research, the objectives, goals, and even achievements of the project to date. You will get a very real sense of how your participation contributes to solving a global research question.

Your assignment is to apply your own skills and talents to the research question. Your support helps to make the project – and over 130 like it – possible. Thank you for contributing your time and money to support scientific research, providing experiential learning, and inspiring environmental responsibility and global citizenship.

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~ *Earthwatch Institute Staff*

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Easter Island Cultures

EXPEDITION BRIEFING

Team I:	September 16 – 29, 2007
Team II:	October 2 – 15, 2007
Team III:	October 17 - 30, 2007

Dear Earthwatch Team Member,

Thank you for taking an interest in the *Easter Island Cultures* project. This long-term study of prehistoric Rapa Nui societal change is entering its 16th year under the Earthwatch banner. Each expedition has produced new discoveries that have substantially increased our understanding of what happened in the past on this tiny speck of land in the middle of the Pacific. This year promises to be the same.

The archaeological record of Rapa Nui (Easter Island) is an impressive one on both large and small scales. The coastal temples (*ahu*) and the statue (*moai*) quarry are places where you will have that "National Geographic moment" upon approaching them for the first time. They are large, impressive and unforgettable. The archaeology of the domestic sphere of life is also rich in detail. The houses, ovens, chicken houses, stone quarries and gardens present on the surface provide us with many clues about how the past society was organized. With a little experience, what will first appear to you as random clusters of unworked stone will soon become recognizable as important features from the past. We are in the process of putting these features together to form a coherent picture of past ways of life.

This year we are working in the A Kivi region on the western coast of the island. Parts of this area have been intensively surveyed by earlier Earthwatch teams and we will continue the survey effort in new and undocumented regions. It is also time to return and evaluate sites through excavation. Specifically, we will be testing numerous household and garden locations to obtain obsidian and carbon for dating so that we can evaluate how the landscape was used in the past. In addition, we will look for fragments of carbonized sweet potato with the hope of solving the long debated issue of when this tuber was introduced. Rapa Nui is a place where almost every shovel of soil yields something, so other surprises may await us as well. As many of you know from past projects, your participation on the expedition is a chance to meet new people, experience a new culture and contribute to a scientific discipline. The *Easter Island Cultures* project offers a lot in each of these areas so please come prepared to work hard and learn about the history and prehistory of this exceptional culture.

Sincerely,

Christopher M. Stevenson, Ph.D.

Easter Island Cultures

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EXPEDITION PACKING CHECKLIST **27**

GENERAL INFORMATION

PRINCIPAL INVESTIGATORS:	1) **Sonia Haoa** 2) **Christopher M. Stevenson, Ph.D.**
POSITION S/TITLES:	1) Independent Researcher 2) Regional Archaeologist
AFFILIATION:	Virginia Department of Historic Resources

PROJECT TITLE:	**Easter Island Cultures:** Changing Agricultural Dynamics on Rapa Nui (Easter Island)
RESEARCH SITE:	Easter Island, Chile
RENDEZVOUS SITE:	Mataveri Airport, Easter Island, Chile
EXPEDITION DATES:	Team I: September 16 – 29, 2007 Team II: October 2 – 15, 2007 Team III: October 17 - 30, 2007
EXPEDITION LENGTH:	14 days
TEAM SIZE MINIMUM/MAXIMUM:	5/7 volunteers
MINIMUM AGE OF PARTICIPATION:	18 years of age *

* It may be possible for 16- and 17-year-olds to participate if accompanied by a parent or guardian. Contact Earthwatch for more information and see Section 9 *'Before You Leave'* for traveling advice for minors.

THE EXPEDITION

1. PROJECT OVERVIEW

The *Easter Island Cultures* research project examines the role of dryland agriculture in the development of social complexity on Rapa Nui (Easter Island), off the coast of Chile in South America. Surface survey and geomorphological techniques are used to identify cultural landscapes and the sites and gardens they contain. Such an approach is significant in that it will identify chronologically early features that will allow the development of a baseline model for Rapa Nui subsistence, technology, and social structure that can be used in the study of internal agricultural development. Earthwatch volunteers will participate in the survey, excavation, and artifact analysis process.

Note: See *Easter Island Cultures: The Research* in the appendix of this briefing for information on the research objectives, methods, and results of this project.

2. RESEARCH AREA

Easter Island is located 3,702 kilometers/2,300 miles west of Santiago, Chile. This triangular island is composed of three volcanoes: Mt. Terevaka, Rano Kau, and Poike. Together they form a landmass of about 105 square kilometers/65 square miles. The terrain between the volcanoes is rolling and contains many smaller cones and basalt flows. The surface is covered with grass, and prehistoric sites can be easily identified from surface remains.

The island's climate is usually pleasant. Daytime temperatures may reach into the mid-90s°F/34-35°C with evening temperatures dropping to the upper 60s°F/20-21°C. On-shore breezes are nearly always present and make working and sleeping very comfortable. However, rain, accompanied by cooler temperatures, may occur for periods of up to four days.

The population of the island (approximately 4,500) consists of local Polynesians and Chilean nationals. Generally people are friendly and the ability to speak Spanish will help you interact. Note that mainland Chile has experienced substantial political change and political discussions are discouraged. Ideological statements contrary to the grain are taken seriously. As a visitor to the island, you should try to remain politically neutral.

Within a short walk of the project headquarters are many points of interest, including caves, petroglyphs, reconstructed and unreconstructed ceremonial centers, swimming areas, and fishing locations. Volunteers should have no problem making use of recreational time. Hanga Roand, the only town on the island, is also nearby and offers restaurants, a post office, a market and various shops. Many of these shops sell wood carvings made by the local people. Simple or sophisticated versions are available and range from US$10 to US$500 in price. Hanga Roand is a dispersed community with houses located on large lots. However, all of the main administrative facilities and major shops are within a half-mile of each other.

3. PRINCIPAL INVESTIGATORS

Sonia Haoa will be present throughout all Earthwatch Expeditions. Dr. Stevenson will be present throughout Team II and will be onsite for half of Teams I and III.

Sonia Haoa C., age 54, is an independent researcher who has had a long association with Earthwatch – this will be her 10th year as Co-Principal Investigator on the *Easter Island Cultures* project. Sonia was born on Rapa Nui, received her B.A. from the University of Chile in 1985, and has participated on scores of foreign research projects over the years. As a result, she is intimately familiar with island archaeology. She and her family run the Hotel O'tai in the center of town, one of the first accommodations of its kind on the island. Sonia speaks English, Spanish and Rapa Nui. She will serve as logistics director, tour guide, artifact analyst, and global positioning system (GPS) expert.

Dr. Christopher M. Stevenson, age 55, is a Regional Archaeologist with the Virginia Department of Historic Resources in Richmond, Virginia. He received his B.A. in Anthropology from Marquette University in 1976, his M.A. in Anthropology from Penn State University in 1979, and his Ph.D. in Anthropology from Penn State in 1984, completing his dissertation on the prehistory of Easter Island. Dr. Stevenson has a devoted interest in the prehistory of Easter Island and this year will represent his 16th field season. He will serve as the overall project director, logistics director, tour guide, lithic analyst, and obsidian hydration dating expert for the project. His hobbies and interests include restoring old houses, stamp collecting, wine making, and fishing.

DAILY LIFE IN THE FIELD

4. VOLUNTEER TRAINING AND ASSIGNMENTS

The Principal Investigators (PIs) of the *Easter Island Cultures* project believe that the key to a successful project is allowing participants to take responsibility and engage in the decision-making process within a structured framework. This approach has worked well for 15 years. During each session crews of three to four persons will be responsible for excavating and recording test units. The same approach applies to the survey process. The PIs will provide guidance and archaeological expertise, while the participants will be responsible for effective organization and data management.

Training

The following topics will be covered through guided tours, informal lectures, and in-the-field training during the course of the expedition. The PI will give the team a more detailed onsite project briefing when you arrive.

- **Easter Island prehistory:** In order to place the research objectives of the project in context, lectures will be presented on the prehistory and ethnohistory of the island. Guided tours will be made of previous excavations and a representative cross-section of island prehistoric sites.
- **Dating methods:** The principles behind radiocarbon and obsidian hydration dating will be presented in the lecture series. Strategies of effective dating will be discussed.
- **Excavation methods:** Project participants will be trained in the techniques used to excavate small structures, open squares, and stratified deposits or middens.
- **Laboratory analysis:** Lab duties may include preliminary artifact sorting and water flotation of soil samples.

Assignments

Project activities consist of landscape survey and site excavation. Participants will specialize in one of these activities for most of the expedition. Water flotation and lab work will occur as needed.

- **Site excavation:** Excavation of the site will entail shoveling, screening, troweling, map drawing, photography, sample selection, and data management. Project members will be divided into teams of three to four persons. Each crew will be assigned to an excavation unit.
- **Field survey**: The field survey team will systematically walk the landscape and record prehistoric sites on the surface. This will entail the measurement of stone features, photography, grass clearing, and Global Positioning System (GPS) data collection.
- **Flotation:** Soil samples will be taken from the many prehistoric pits or midden deposits encountered during excavation. In order to recover small botanical and faunal remains, water flotation techniques will be used. Flotation will occur when a sufficient number of samples have been acquired for a full day's work.
- **Lab work:** Lab days will be scheduled as needed and may occur on rainy days. Toward the end of the project a portion of the field crew may be assigned to the lab. Duties will include artifact washing and sorting, lithic analysis, and flotation of carbonized plant remains.

5. Team Itinerary

Below is a tentative itinerary for the expedition. The order of specific events may vary due to changing flight schedules, weather conditions, etc.

Day 1:	Rendezvous, settle into accommodations and meet crew members and fellow participants, evening orientation session
Day 2:	Orientation and instruction in the project methods, tour of the survey and excavation areas
Days 3-6:	Field excavation, lecture or recreational time in the evenings
Day 7:	Archaeological tour of the ceremonial center of Orongo and west coast archaeological sites, including Cannibal Cave, Vinapu, and a museum
Day 8:	Archaeological tour of the Rano Raraku crater and statue quarry and south coast *ahu*, followed by a trip to the north coast beach
Days 9-15:	Field excavation and lab work, lecture or recreational/rest time in the evenings
Day 16:	Depart from Easter Island

6. Daily Schedule and Tasks

A typical research day is outlined below. Please be aware that schedules can and do fluctuate due to weather, research needs, etc. Your cooperation and understanding are appreciated. On Days 7-8 the team will tour the area rather than conduct research work, and breakfast will be an hour later.

7:30 am:	Breakfast, prepare for the field
8:00 am:	Leave for the field or lab
8:30 am:	Begin fieldwork
12:00 pm:	Lunch
12:30 pm:	Continue fieldwork
5:00 pm:	Wash up, dinner
8:00 pm:	Organizational meeting and/or lecture (when scheduled)
9:00 pm:	Recreational/Rest time

7. Accommodations

Team members will be housed near the town center of Hanga Roa at Hotel O'tai, a small family run enterprise. The hotel has very pleasant dining rooms and is surrounded by a garden area. It is just a three-minute walk to the harbor, shopping, and statues. Project staff will stay in a house nearby. The team will be transported to the field from the accommodations in a large van and/or truck.

Each room at the Hotel O'tai will be occupied by two people, so volunteers who join the expedition together will be able to share. Smokers will be separated from non-smokers when possible. Each room has a bathroom with a flush toilet and hot shower. However, not everything works perfectly on Easter Island and you should expect to a few cold showers. This also applies to other aspects of life on the island. The hotel is comfortable and clean, but expect a cockroach or two at night as they are found everywhere despite efforts to keep them out.

The hotel will furnish sheets, blankets, and towels. Participants should bring all necessary toilet articles. For a small charge it is possible to have your laundry collected from your room and done on the premises. The electricity in the rooms is 220-volt and requires a converter, so electronic devices are not recommended. **Please note that propane, electricity, and water are limited and valuable resources on Easter Island, so volunteers are asked to conserve.**

There will not be email access at the hotel but there are several internet cafes within a few minutes' walk. Phone calls can be made by purchasing a phone card at the grocery store and calling from a payphone. Phone cards can also be purchased at a telephone office downtown, from where international calls can be made.

8. Food

Breakfasts and dinners will be eaten together with staff at the hotel. The composition of the evening meal will depend on the daily fish catch and what came in on the plane the day before. Bag lunches will be provided on working days. The hotel cooks are very creative and the meals are tasty. However, certain items may not be available on the island, so **if you wish to have specific food items/snacks, you must bring them with you**; you are also encouraged to bring something for the group to share to help boost team morale. Beef jerky, granola bars, chocolate, ground coffee, wine, and scotch have proven to be favorite items.

Below are examples of the foods and drinks you might expect during your expedition. Please bear in mind that variety depends on availability. This list is intended to provide a general idea of food types, but it is very important that volunteers be flexible.

Breakfast:	Bread, honey, margarine, eggs (pound cake, papaya and other fruits may be available at times)
Lunch:	Sandwiches, pasta, eggs, chicken, fruit
Dinner:	Meat, fish, salad, taro, potato, bread, canned or fresh fruit, dessert
Snacks/Other:	Bring along a supply of whatever you might need/want (some junk food, such as chips and candy, is available but may be expensive and damaged after the long flight to the island)
Beverages:	Water (the water supply is treated and safe to drink), instant coffee, tea, juice from powdered mix (beer, wine and liquor are also available for purchase)

When the schedule permits, team members who wish to eat out may visit one of the restaurants in town. The local restaurants are pleasant places to eat and drink and may have items that are not available at the stores. A good meal may cost about US$15-20.

Special Dietary Requirements

Please alert your Earthwatch Expedition Coordinator to any special dietary requirements as soon as possible (e.g. diabetic, lactose intolerant, etc.). No individual meals will be prepared and flexibility is a must. If you need certain food items, please bring them along.

Be aware that the diet on Easter Island is not appropriate for strict vegetarians or vegans. It may be possible to get meatless meals but you may have a problem avoiding animal products altogether. If this poses a problem, then participation on this Earthwatch Expedition should be seriously reconsidered.

TRAVEL PLANNING

9. BEFORE YOU LEAVE

For a listing of useful websites for passport and visa requirements see Section 15 *'Helpful Resources.'*

Entry Information for Chile

All volunteers should check with a travel or visa agency for specific visa and entry requirements for Chile. Citizens of the US, UK, EU, Japan, Australia, and New Zealand will only need a passport valid for at least six months beyond the dates of travel, but please check with a travel agent for changes. For US citizens there is a US$100 fee (good for the life of the passport) required to enter Chile that is paid at the Customs station in Santiago. Citizens of other countries may be charged similar fees (see http://www.wordtravels.com/Travelguide/Countries/Chile/Visa).

Essential Information if You Need a Visa to Enter Chili

Type of Visa	You must get a **TOURIST VISA**.
Where to Get a Visa	Contact the nearest **Chilean Embassy or Consulate** to find out how to apply for your visa. Please note that this process can take weeks or more. If you have less than six weeks or wish to save yourself trouble, we strongly recommend using a **visa agency**, which can both expedite and simplify the process.
Required Information	You will need to send your **passport** (valid for at least six months beyond your stay), a **Visa Application and Immigration Form, 2-4 passport-size photos plus payment** to the embassy or visa agency (if applicable). Please be sure that your passport is valid for at least six months beyond your stay.
Contact Information	You may be required to list the following contact information on your Visa Application and Immigration Form: **Hotel O'tai** Isla de Pascua, Chile Tel: +56-32-1-100-250
Cost of a Visa	Generally between US$40-100, but varies from country to country and can potentially cost **up to US$180**. A visa agency will charge an additional fee (depending on the amount of time it takes to process the application), which you can inquire about directly.

Reminder: The purpose of your visit is for vacation, holiday or travel. Foreign immigration officials do not always understand the concept of a "working vacation" or even "volunteering." Words such as "working"/"volunteering," "research" or a "scientific expedition" can raise questions concerning the country's foreign labor laws and/or prompt questions about official scientific research permits and credentials, etc., to which volunteers on their own will not be equipped to respond. All required research permits for the project are in place and have been approved by the proper authorities.

Volunteers Under 18 Years of Age

Entry to Foreign Countries

In an effort to prevent international child abduction many governments have initiated procedures at entry/exit points. It may be possible for 16- and 17-year-olds to participate in the project if accompanied by a parent or guardian. In this case, if the minor will be traveling with only one guardian or if for any reason they will be traveling alone, it may be necessary to have a notarized letter from all legal guardians stipulating that they may travel unaccompanied or in the presence of a single guardian. This letter must give an explanation for why only one parent or someone other than a parent is signing the letter. For example, if one parent is deceased, only one parent has legal guardianship, or someone other than the parents are legal guardians, the letter should state that.

In addition, airlines may also have documentation requirements for unaccompanied minors. Parents of minors are responsible for checking with each airline that their child will be flying to ensure that sufficient documentation is provided. This could include a copy of a birth certificate or a notarized letter stating that the minor has his or her parent's permission to travel alone or with only one parent.

Note: Requirements by specific countries and airlines vary and change frequently. You MUST keep informed of the requirements on your own to avoid problems at immigration. If a letter is not available, the volunteer under 18 can be refused entry into the country. There is nothing Earthwatch Institute can do to help in this circumstance.

Travel Medical Insurance

Travel medical and evacuation insurance is mandatory for Earthwatch volunteers while on an Earthwatch expedition anywhere in the world. The cost of this insurance is included in your volunteer contribution. It covers volunteer travel medical risk, including medical expenses and medical evacuation, while traveling with Earthwatch overseas or on an expedition within your home country. Without insurance, the costs of such measures can range from US$20,000 to $50,000.

The emergency medical and evacuation assistance provider for Earthwatch is On Call International. On Call is a 24-hour international operation which provides medical assistance and evacuation, a 24-hour nurse help line and other travel assistance services such as lost baggage and lost document assistance.

Basic coverage is valid in the country of your Earthwatch expedition and during international travel to and from your expedition. If the expedition takes place in your home country, coverage begins when your group forms for the expedition and ends when the group disbands, and is incremental to your existing health insurance. Options are available for volunteers who would like to extend the period of coverage, increase insurance amounts or purchase additional cancellation or baggage insurance. Application forms for additional coverage are included in your volunteer packet.

A detailed description of the Volunteer Medical and Evacuation Insurance Program policy, including the optional coverage increases, will be sent with this briefing. **Please note that policies are specific to each Earthwatch office.**

To contact On Call International in the event of an emergency, dial:

- 1-866-509-7715 from within the US
- +1-603-898-9159 from outside the US

State that you are on an Earthwatch expedition. The Earthwatch policy number is #US008020.

Cancellation Insurance

Trip cancellation insurance is highly recommended for Earthwatch volunteers. Depending on the level of coverage you purchase, cancellation insurance will help cover your airfare and Earthwatch contribution if you need to cancel your expedition due to medical or other covered reasons. Earthwatch Institute does not reimburse airfare or costs associated with cancelled flights or expeditions. Levels of reimbursement for cancelled airline tickets or ticket change fees will vary depending on what type of trip cancellation policy you purchase. You are strongly advised to buy flexible or refundable plane tickets. Note that volunteers with preexisting medical conditions are encouraged to explore their coverage options.

For US and Canadian Volunteers

Earthwatch is offering comprehensive optional travel insurance through CSA Travel Protection as a service to our US and Canadian volunteers. While our inclusive insurance covers your emergency medical needs while in the field, this optional policy covers trip cancellation insurance due to medical emergencies, lost luggage, travel delays, etc. For more information on the insurance policy, call Earthwatch at 1-800-776-0188 or visit www.csatravelprotection.com. Please note that some coverage is dependent on purchasing insurance within 24 hours of paying in full for your expedition. Should you decide to take out our optional insurance, please use the following producer code to indicate your affiliation with Earthwatch: 83534816.

For Volunteers Signing Up through Earthwatch Europe

Earthwatch Europe volunteers can purchase travel insurance from Earthwatch that is underwritten by Endsleigh and includes Additional Cancellation Cover. Additional Cancellation Cover insurance includes cover for non-refundable travel expenses should your expedition be cancelled. Alternatively, if Earthwatch Europe volunteers hold their own travel insurance they may be able to purchase Additional Cancellation Cover through their existing insurer.

Travel Agencies

Contact your local travel agent or use the web to find the lowest rates to make your travel arrangements. A list of suggested travel agents can be found in Section 15 *'Helpful Resources.'* Be sure to give your rendezvous details to your travel agent as soon as possible so they can plan your trip accordingly.

Other Advice / Information

- *Local currency:* Chilean peso
- *Languages:* Spanish and Rapa Nui (English will be the official project language)
- *International dialing code:* +56
- *Electricity:* 220 volts, 50 hertz, plug types C and L
- *Time zone:* Six hours behind Greenwich Mean Time (GMT/UTC-6)
- *Personal funds:* You are advised to change a good sum of money in the Santiago airport prior to going through Customs, though US dollars are readily accepted on the island. Traveler's checks are accepted at larger stores and hotels, but are generally not advised. There is one cash machine/ATM but it may not work. There will be no opportunities to go to a bank during the expedition. **Credit cards are rarely accepted on the island.** Tipping of taxi drivers and restaurant waitstaff is not generally practiced on Easter Island. Additional expenses mainly include the costs of entertainment and touring during recreational time.

- *Estimated additional costs*:
  - National Park fee — US$15
  - Entry fee at Customs — US$100 (one time fee in Santiago for US volunteers)
  - Museum entrance fee — US$3
  - Laundry* — US$30
  - Car rental* — US$60 (8 hours)
  - Wine Club* — US$40
  - Horses* — US$25 (8 hours)
  - Shopping* — US$100 (souvenirs, woodcarvings, etc.)
  - Meals out, etc.* — US$60

  *Optional expenses
- *Luggage storage*: Luggage may be stored at Security on the first floor of the international airport in Santiago, Chile.

10. Project Conditions

Please show this section to your physician when he/she is completing your health statement. Be sure to discuss inoculation requirements with your physician well in advance of your departure date. See Section 11 'Health Information' for inoculation information.

To the examining physician:

Your patient has volunteered to join a field research team that has specific physical demands of which you and your patient should be aware. **We need your accurate evaluation of your patient's ability to meet the conditions detailed below in order to safeguard his/her health and safety and ensure that he/she can participate fully and effectively.**

General Conditions of the Research Site

Easter Island has a temperate climate. The expeditions will take place during the spring season, before the intense heat of summer sets in. The weather may range from cool and rainy to relatively hot and sunny. Warmer temperatures will predominate. Below are average conditions for the expedition period.

Humidity	60%	to	70%
Temperature range	60°F/16°C	to	90°F/32°C
Altitude	0 ft/m	to	1,200 ft/400 m
Rainfall	0-1 in/20 mm	per	week

Physical Demands

All team members should be in good physical condition. Site excavation and survey require a considerable amount of physical exertion in the form of walking, kneeling, bending, shoveling, and screening. Some terrain is difficult for walking. There will be regular breaks but up to 4-5 hours per day may be spent on these physical activities. It will be necessary to carry the excavation trunk and shovels to the site each day, but it will not likely be necessary to transport heavy equipment. People who do not have serious health problems can effectively participate in the excavation, as long as they have fairly strong knees and backs and are otherwise physically fit. Volunteers with certain physical limitations may be able to participate in various project activities.

Team members will also participate in landscape surveys during part of the expedition, which require extensive walking over stones and lava flows for up to six hours a day. The walking will be

done at an even and slower pace but still requires considerable agility. Participants should be able to walk 3-4 miles/5-6.5 kilometers per day over rugged terrain, a portion of which will be on an uphill grade.

Below are the expected demands of the project, but please keep in mind that conditions may change and the project could potentially be more or less strenuous than the chart indicates.

Activity	**Workload/Intensity**
Sitting	0.5-1 hours per day
Bending	2.5-4 hours per day
Walking	4-5 hours per day
Carrying	10 lbs/4.5 kg for 0.25 hours per day
Digging	4-5 hours per day

Potential Hazards

Hazard Type	**Associated Risks and Precautions**
Terrain	The terrain can be rough and uneven and may include steep drop-offs and vertical caves hidden by vegetation. Stones and depressions are hidden in the grass and may cause scrapes or twisted ankles. High cliffs surround much of the island, including around archeological sites the team will visit and near the survey area. The most common injuries are a result of walking over the rough terrain and on dirt piles at the site, though these are generally minor.
Wasps and scorpions	Wasps are rarely encountered and scorpions may be found under rocks at the site. Both sting and leave painful welts, but these do not require medical attention unless there is a strong allergic reaction. Stings can be avoided with caution.
Dengue fever	A less severe form of Dengue fever appeared on the island in February 2001. This illness is transported by mosquitoes that are present in the evening and night hours. There is no direct cure for the illness other than bed rest, non-aspirin based pain relievers and fluids. The best prevention is wearing insect repellent.
Climate/ Weather	Volunteers should take appropriate precautions to avoid dehydration and overexposure to the sun. It is essential to wear sunscreen and drink fluids regularly.
Swimming	Swimming areas are in shallow water but mild undertows can be present.
Social concerns	Youth alcoholism is a local problem. Please do not accept invitations to parties.
Project equipment	Sharp excavation equipment, such as trowels and shovels, is used and can cause injury without proper handling.
Security	Avoid secondary streets with limited lighting at night and do not walk alone.
Dogs	Dog packs form in the evenings on the streets and are potentially dangerous. Please do not approach these animals.

Medical Conditions of Special Concern

Condition	**Concerns and Precautions**
Lack of physical fitness	Some degree of physical stamina is required for excavation and survey. Volunteers with certain physical limitations may be able to participate in alternative activities, but in general people who are not physically fit should reconsider participation.
Bone or joint problems	Strong knees and backs are important. Volunteers with bad back, knees, ankles, etc. would have difficulty participating.
Any condition requiring immediate specialty care	Limited medical services are available on the island; the nearest full-facility hospital is in Santiago. Depending on flight schedules, it may be a couple of days before transport to the city would be possible, so this project is not appropriate for those with chronic conditions (e.g. heart or kidney problems, sever asthma).

11. HEALTH INFORMATION

All volunteers should make sure to have the following up-to-date immunizations: DPT (diphtheria, pertussis, tetanus), polio, MMR (measles, mumps, rubella) and varicella (if you have not already had chicken pox). If coming from a country or region where yellow fever is endemic, a certificate of vaccination will be required.

Medical decisions are the responsibility of each volunteer. Note that health conditions around the world are constantly changing, so keep informed and consult your physician, a local travel health clinic, the US Center for Disease Control (www.cdc.gov), the World Health Organization (www.who.int) or the resources in Section 15 *'Helpful Resources'* for the latest health information for travelers. Please consult your physician for guidance on inoculations if you intend to travel to other parts of the country.

12. PACKING CONSIDERATIONS

PLEASE SEE THE PACKING CHECKLIST AT THE BACK OF THIS BRIEFING AND REMEMBER TO TAKE YOUR BRIEFING WITH YOU ON YOUR EXPEDITION.

General Considerations

International flights usually restrict passengers to two pieces of luggage (88 pounds/40 kilograms) and one carry-on. Luggage with wheels is appropriate, but do not bring more luggage than you can carry and handle on your own. Remember to pack a carry-on bag with an extra set of field clothing and personal essentials in case your luggage is lost and/or takes several days to catch up with you.

Dress on the island is casual. For fieldwork, bring durable clothing including jeans or work pants, work boots, and strong gloves to handle coarse basalt. For evenings and recreational days, bring shorts, a swimsuit, and comfortable shoes. Bedding will be provided by the hotel, but bring an extra towel for the beach. Remember to bring a full supply of any necessary prescription medicines.

Weather Considerations

Please take weather conditions into consideration when packing for your expedition. Working conditions on the island may range from very pleasant to miserable. The days will likely vary from warm to hot (70–90°F/21-32°C). Note that while the continuous breeze may make it feel cool, the solar radiation is very strong. Bring lightweight clothing, a hat and sunscreen. It will also rain at times so rain gear is advised. Climate information can be found in Section 10 *'Project Conditions.'*

Cultural Considerations

If you wish to go to Catholic mass on Sunday a nicer set of clothes is appropriate. It is also appropriate to take a few minutes to learn the local greetings in either Spanish or Rapa Nui. As mentioned in Section 2 *'Research Area,'* political discussions are discouraged.

Essential Items

Make sure to bring your Earthwatch Expedition Briefing with you! It includes essential information to which you may need to refer during your expedition, as well as during your journey to and from the rendezvous site. **Please see the Expedition Packing Checklist for a complete list of what you will need to take with you.** You are encouraged to go through the list and mark off each required item right before you leave for your expedition.

13. RECOMMENDED READING

Please read *Easter Island Cultures: The Research* in the appendix of this briefing. This document was prepared by the Principal Investigators and Earthwatch and explains the research conducted through this project as well as some results to date. You are also requested to read the following article, which can be found online at http://www.earthwatch2.org/secondary_expedition_pages/briefings/stevenson_addendum1.pdf:

- Stevenson, C., Jackson, T., Mieth, A., Bork, H. and Ladefoged, T. 2006. Prehistoric and early historic agriculture at Maunga Orito, Easter Island (Rapa Nui), Chili. *Antiquity* 80:919-936.

Below are additional recommended materials for those interested in further preparing for the expedition. Many can be purchased online through popular vendors. See Section 15 *'Helpful Resources'* for suggested vendor websites.

- Hunt, T.L. and Lipo, C.P. 2006. Late Colonization of Easter Island. *Science* 311:1603-1606.
- Bahn, P. and Flenley, J. 1992. *Easter Island, Earth Island*. Thames and Hudson, Ltd., London.
- Lee, G. 1990. *An Uncommon Guide to Easter Island (*a unique guide to Easter Island with answers to many questions; available at bookstores and from International Resources, P.O. Box 840, Arroyo Grande, CA 93421-0840, for US$18.95 plus shipping).
- Metraux, A. 1940. *Ethnology of Easter Island*. Bernice P. Bishop Museum, Bulletin No. 160. Honolulu (comprehensive description of Easter Island material culture and customs).
- Rainbird, P. 2002. A Message for Our Future? The Rapa Nui (Easter Island) Ecodisaster and Pacific Island Environments. *World Archaeology* 33(3):436-451.
- Routledge, C. 1919. *The Mystery of Easter Island*. Sifton Praed, London (description of her long stay on the island in the early 1900s with information on island archaeology and customs).
- Stevenson, C. 1986. The Socio-political Structure of the Southern Coastal Area of Easter Island, In: *The Evolution of Island Societies*. Patrick V. Kirch, editor. Cambridge University Press (discussion of the types of religious ceremonial centers, what they meant in social terms, and how they changed over time).
- Stevenson, C and Ayres, W.S. 2000. *Easter Island Archaeology: Research on Early Rapa Nui Culture*. Easter Island Foundation, Los Osos, CA (available from EIR, P.O. Box 6774, Los Osos, CA 93412-6774 for US$25 plus shipping).
- Van Tilburg, J. 1994. *Easter Island: Archaeology, Ecology and Culture*. British Museum Press, London.

Project Field Report

Each Earthwatch Institute-supported project submits a report on the past year's research and results to Earthwatch, generally on an annual basis. The most recent field report for this project may be available online through www.earthwatch.org/expeditions/stevenson.html. Note that reports are not available for all projects.

14. EMERGENCIES IN THE FIELD

The following moderately equipped hospital and dental facility is located one mile/1.6 kilometers from town and 5-8 miles/8-13 kilometers from the work site, though the trip from the work site is on slow/rough roads and can take up to three hours:

- Hospital Hanga Roa
  Isla de Pascua, Chile
  Tel: +56 32-2-100-215
  Distance: 5-8 miles/8-13 kilometers or 1-3 hours

There is a commercial flight to the mainland twice a week for transport to Santiago, where a full-facility hospital is located. Transport to Santiago in the event of an emergency might take up to three days depending on flight times.

15. HELPFUL RESOURCES

Passport and Visa Information

- Embassies around the world: http://www.embassyworld.com
- For Japanese citizens: http://www.rainbowt.jp/travel/visa_top.html
- For Australian citizens: https://www.passports.gov.au and http://www.dfat.gov.au/visas/index.html
- Passport Visa Express (for US citizens): www.passportvisasexpress.com
- The Visaservice: http://www.visaservice.co.uk
- Thames Consular Services Ltd: http://www.visapassport.com

Travel Guidebooks and Booksellers

- Lonely Planet: http://www.lonelyplanet.com
- Rough Guide: http://travel.roughguides.com
- Amazon: http://www.amazon.com
- Barnes and Noble: http://www.bn.com

Travel Agencies and Advice

- Exito Travel (familiar with Earthwatch and specializes in Latin America):
  http://www.exitotravel.com
  108 Rutgers St.
  Ft. Collins, CO 80525
  Tel: 800 655 4053, ext. 8507 (toll free US and Canada, ask for Isaac Hilpman)
  Email: isaac@exitotravel.com
  Fax: +1 510 868-8306 (worldwide)
  Code: EARTH
- STA Travel (contact Angie Kurtz or Chris Chappell and mention that you will be going on an Earthwatch Expedition): http://www.statravel.com
  36 Geary Street
  San Francisco, CA 94108
  Tel: +1 415 391-8407
  Email: sfo@statravel.com

- STA Travel (UK): http://www.statravel.co.uk
  Tel: +44 (0) 1865 792800
  Fax: +44 (0) 1865 792911
  Email: manager.oxford@statravel.co.uk
  Quote code: EWE01/02
- Wexas International (Europe): http://www.wexas.com
  Tel: +44 (0) 20 7581 8761
  Fax: +44 (0) 20 7581 7679
  Email: southern@wexas.com
  Quote code: EWE01/02
- Democracy Travel (contact Jean S. West, Assistant Manager)
  4818 MacArthur Blvd NW
  Washington DC 20007
  Tel: 202 965 7200 or 866-557-9968 (toll free US and Canada)
  Fax: 202 342 0471
  Email: jean@democracytravel.com
- World Travel Guide: http://www.worldtravelguide.com
- UK Foreign Office travel advice: http://www.fco.gov.uk/travel
- Third World Traveler: http://www.thirdworldtraveler.com/Travel/Travel_Links.html

Airline/Airport Resources

- Flight comparison tools: http://www.bookingbuddy.com and http://www.1800-fly.com
- Airport codes worldwide: http://www.logisticsworld.com/airports.asp

Country Information

- Country information from around the world: http://www.countryreports.org
- National Geographic Map Machine: http://plasma.nationalgeographic.com/mapmachine
- US State Department: http://www.state.gov
- Time worldwide with GMT/UTC: http://www.worldtimeserver.com
- Currency converter: http://www.xe.com
- Electrical current converter: http://kropla.com/electric2.htm
- Telephone dialing codes: http://kropla.com/dialcode.htm
- Online unit conversions: http://www.onlineconversion.com
- Worldwide weather: http://www.wunderground.com or http://www.tutiempo.net/en
- ATM locator: http://visa.via.infonow.net/locator/global/jsp/SearchPage.jsp or http://www.mastercard.com/atmlocator/index.jsp

Health Information

- Travel health website: http://www.mdtravelhealth.com
- Center for Disease Control: http://www.cdc.gov
  Tel: +1 800 311-3435 or +1 888 232-3228
- World Health Organization: http://www.who.int
- The Travel Doctor: http://www.tmvc.com.au
- Disease outbreaks: http://www.who.int/csr/don/en
- Hospital for Tropical Diseases: http://www.thehtd.org
- Travellers Healthline Advisory Service
  Tel: 020 7950 7799
- MASTA Travelers' Healthline (UK)
  Tel: 0906 8 224100 (within UK)

APPENDIX

EASTER ISLAND CULTURES: THE RESEARCH

The following information was taken from the research proposal submitted by the Principal Investigators to Earthwatch Institute. Included is a description of the research conducted through this project, some results to date, and other information regarding the accomplishments of the project and the staff. Specific details regarding research sites, methods, etc. is subject to change slightly from year to year and such changes may not be incorporated into this document.

BACKGROUND, OBJECTIVES, AND METHODS

Introduction

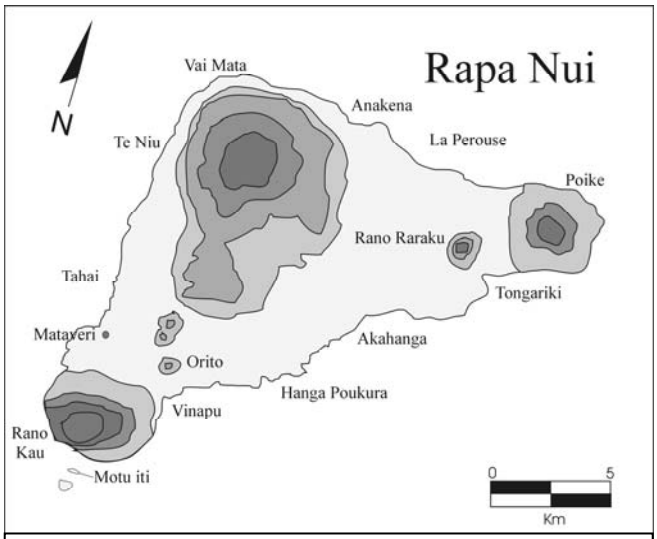

Figure 1: The island of Rapa Nui

Easter Island, or Rapa Nui, is often depicted as a microcosm for global environmental disaster (Diamond 2005; Kirch 2006). It has been hypothesized that rapid population growth initiated landscape degradation, which in turn resulted in a devastating societal "collapse" (Kirch 1984, 2000; Bahn and Flenley 1992; Flenley and Bahn 2002; Diamond 2005). Agricultural development has been implicated in the clearance of native vegetation, and this deforestation is thought to have loosened the soils and enhanced physical erosion (Mieth and Bork 2004). While physical erosion might have been significant in some areas of the island, recent research in Hawai'i suggests that agriculture can also deplete the inherent nutrient properties of soils (Vitousek *et al.* 2004; Hartshorn *et al.* 2006; Meyer *et al.* 2007). If soils are relatively old and receive high rainfall, they have little nutrient reserve and hence depletion can occur quickly when humans use the land. The Hawai'i research has defined the geologic age and rainfall levels under which soils are depleted and could not have sustained intensive dryland agriculture. While there were a range of environmental zones in Hawai'i where alternative agricultural strategies could have been used (e.g. pondfielding in windward wetland areas), in Rapa Nui there were fewer options. Agricultural development in the dynamic Rapa Nui environment demanded significant innovations, and the unique strategies of lithic mulching and rock gardens were employed.

This research project examines the role of prehistoric lithic mulching and rock placement in Rapa Nui gardens and agricultural systems. It is postulated that these anthropogenic techniques were developed in response to the naturally poor soils of the island and the effects of environmental degradation. Archaeological studies suggest that rock based gardening provided most of the nutritional requirements of people and surpluses that funded activities such as the construction of monumental architecture (Stevenson *et al.* 1999, 2006; Wozniak 1998). We suggest that the

development of rock gardens and mulching originated in response to a number of factors that included: 1) an island-wide distribution of nutrient poor soils that resulted from relatively old volcanic substrates and high levels of rainfall leaching, 2) an increasingly poor environment caused by deforestation and a rapid decline of organic inputs into the soil, and 3) the exposure of the deforested landscape to higher wind velocities, temperature extremes, and possibly physical erosion.

Our research will focus on key environmental and archaeological data that will allow us to determine how the population responded to a series of challenging environmental problems in developing agricultural resources. We will examine the dynamic nature of the island's soil nutrients, the extent of physical erosion, the negative effects that agricultural might have had on soil nutrients, and how lithic mulching might have been developed to enhance nutrient levels. The process of agricultural development was not simple, but varied across the landscape with elevation, orientation, soil type, and rainfall amounts. As a result, it is important to empirically assess the differences in farming technologies in response to these key variables.

Rapa Nui Agricultural Production

Throughout Polynesia terrestrial resources were grown in both wet and dry locations (Kirch 1994). On Rapa Nui, the lack of permanent streams restricted agricultural activities to dryland contexts, with limited water diversion (Stevenson 1997). A range of dryland cultigens were grown, the most important staples being dryland taro, yams, and sweet potato (Flenley 1993). The methods used to grow these cultigens were not static, rather developed throughout centuries of occupation.

On Rapa Nui the trajectory of agricultural development is not well established. However, a number of studies (Mieth and Bork 2003; Stevenson 1997; Stevenson *et al.* 1999, 2006; Wozniak 1999) have begun to define the broad outlines of change. Prior to human occupation, a mature forest covered most of the island. The palm trees and the other half-dozen hardwood species (Orliac and Orliac 1998) provided protection against the nearly constant onshore winds and maintained nutrients in the soil through the decay of leaf litter. Orographic rainfall on Rano Kau and Mt. Terevaka volcanoes is an important variable for vegetative growth (Benedetti 1991). As elevation increases so does annual rainfall which is estimated to have been significantly higher than that recorded for lowland settings near the current town of Hanga Roa (i.e. 1,100 millimeters per year). However, as elevation increases so do wind velocities, which may be up to 0.5 meters per second greater for each 100 meters of elevation above coastal elevations. These velocities can increase evapotranspiration and damage the leaves of unprotected plants. Cooling with elevation also occurs and in conjunction with the other factors reduces the chances of plants achieving maturity.

An additional limiting factor for vegetative growth was the nutrient poor soils that covered much of the island. The geological substrates primarily consist of older soils in excess of 3,000,000-190,000 years (Gonzalez-Ferran 2004) that have been leached of their nutrients by rainfall over the millennia. These soils are low in plant available phosphorus (P) as a result of their andic properties (Louwagie *et al.* 2006; Ladefoged *et al.* 2005). Low P availability can reduce the overall productivity by preventing full plant development. Although much of the island contains relatively poor soils, there are areas that contain richer soils as a result of more recent lava flows and ash deposits.

The initial settlers of Rapa Nui faced these environmental constraints with a transported agricultural economy reliant upon the tubers of yam (*Dioscorea alata*) and taro (*Colocasia esculenta*), and supplemented by sugarcane, bananas, bottle gourd, and other minor plants (Flenley 1993). At some point in prehistory the sweet potato (*Ipomoea batatas*) was introduced, and Wallin *et al.* (2005) suggest that this occurred around the 14th century and provided a valuable new cultigen. The transferal of these plants to a far southern latitude meant that climatic conditions were less than optimal for the yam and taro, which were near their lower tolerances with regard to moisture and

temperature (Louwagie *et al.* 2006). The sweet potato, however, could mature under these cooler and dryer conditions within shallow soils characteristic for many parts of Rapa Nui.

During the first few centuries of occupation Rapa Nui was deforested and transformed into a landscape consisting primarily of gardens, grass, and ferns. Some researchers (Flenley and Bahn 2002) have suggested that this deforestation was the result of human actions, whereas others (Hunt 2007) have proposed that introduced Polynesian rats (*Rattus exulans*) accelerated deforestation by killing seeds and thus interfering with the natural regeneration process. The evidence generally suggests that trees may have existed at higher elevations until ca. AD 1500 but these were also eventually removed (Mieth *et al.* 2002). Wood from many of the trees would have been used as fuel, for the construction of housing, tools, and small boats, as well as for the transportation of monolithic statues (*moai*). However, the need for expanded agricultural plots to meet a growing population most likely initiated larger scale clearance of forests. Fire was the primary method for removing trees and these clearing events have been identified in stratigraphic profiles at Te Niu on the west coast and on the Poike peninsula that date to around AD 1300 and AD 1400, respectively (Horrocks and Wozniak 2007; Mieth *et al.* 2002).

Forest clearance opened land up for agricultural activities and increased the annual productivity of the landscape. In the short term the amount of foods could be increased, however, perhaps unanticipated by the early farmers was the long-term effect of forest clearance on the agricultural system. We can identify four detrimental changes that would have negatively impacted productivity using traditional open-field farming practices:

1) Increased wind velocities that damage the leafy portion of plants
2) Increased average surface temperatures and thermal extremes in the upper soil horizon that would increase plant moisture requirements
3) The elimination of organics entering the soil from plant litter
4) Higher wind velocities and direct sunlight that reduce the amount of available moisture in the upper soil horizon

Prehistoric Rapanui farmers apparently realized the detrimental impacts of deforestation and responded with the development of rock gardens constructed from basalt stones and boulders found at the numerous surface lava flows on the island landscape. Data pertinent to dating the age of this innovation are few. Investigations within a rock garden at Mt. Orito (Figure 1) by Stevenson *et al.* (2006) tentatively suggest that rock-free gardens were abandoned in the 15th century. We propose that the transition to rock garden agriculture occurred around this time and was rapid once the beneficial effects of the rocks were understood.

Rock gardens are discreet architectural entities that were landesque capital improvements. Survey of the Rapa Nui landscape has resulted in the identification of four rock garden forms (Stevenson *et al.* 1999; Stevenson *et al.* 2006):

1) Lithic mulch gardens that consist of a dense surface occurrence of pre-selected stone 5-12 centimeters in diameter underlain by high densities of rock that has been worked into the top 20-30 centimeters of the agricultural (Ap) horizon
2) Scree pavements comprised of a single layer of surface stone that forms of near continuous distribution; few stones are present in the Ap-horizon
3) Boulder gardens formed from basalt blocks 30-80 centimeters in diameter that were intentionally placed on the surface of the Ap-horizon
4) Garden depressions in deep (over 50 centimeters) lithic concentrations found at the base of hills and inflections in the landscape

Boulders are also found in conjunction with lithic mulch gardens and scree pavements to form other configurations.

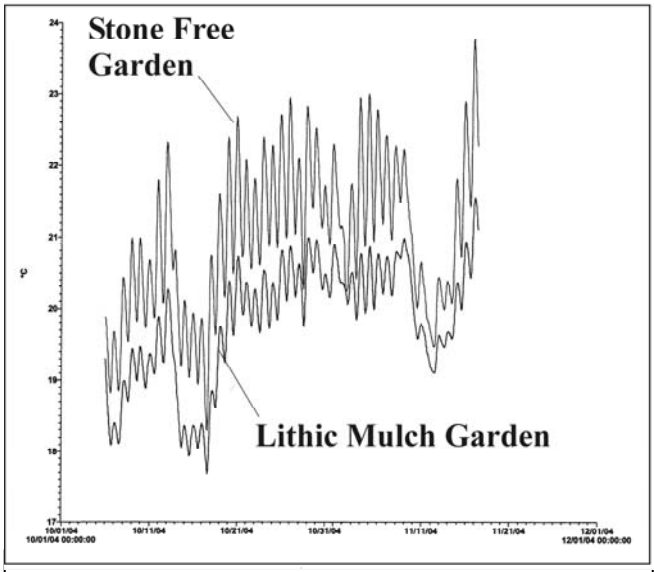

Figure 2: Diurnal temperature profiles under lithic mulch (lower profile) and in a stone free garden (upper profile)

The intentional placement of rock within and on top of the Ap-horizon might have provided three important benefits: 1) boulders could have provided protection from wind, 2) closely spaced surface rock might have promoted moisture percolation and prevented evaporation, and 2) lithic mulch might have served to reduce the surface temperature extremes that resulted in a lower average temperature for soils beneath the rock cap. Figure 2 shows the results of initial thermal monitoring on Rapa Nui for a three-month period in 2004. The graph shows the lower amplitude of the diurnal temperature for a lithic mulch garden in relation to an adjacent stone free garden. These three possible benefits of rock placement would have reduced the environmental stresses on agricultural plants and fostered their growth to full maturity.

Rapanui farmers also used natural landscape features to further protect agricultural fields from environmental forces. Systematic survey has tentatively noted that gardens are concentrated in undulating terrain within swales and at the bases of hills (Stevenson and Haoa 2007). Even small inflections on flat terrain forms appear to have been used to divert air flow over the plants. Gardens also tend to be concentrated at the base of basalt outcrops that not only offered wind protection, but also provided a source of raw material for boulders and lithic mulch. Rock fracture patterns clearly show that outcrops were systematically pried apart along planes of weakness and the boulders were then moved into the garden plots.

It is our hypothesis that prehistoric Rapanui were able to sustain high levels of production under marginal conditions not only by cultivation with lithic mulch, but also by matching rock garden types with specific climatic, physiographic, and soil nutritional settings on the island landscape. For example, we have noted that lithic mulch gardens tend to be present on shallow soils where the decomposing parent material is within 20-40 centimeters of the surface, while scree pavements and boulder gardens are located on deeper soils that have deeper pre-settlement colluvial deposits. We hypothesize that an important aspect of maintaining high productivity was the ability to minimize unnecessary expenditures of effort and this could be accomplished by selecting the most appropriate crop and garden configuration for specific site conditions.

We also hypothesize that the use of lithic mulch technology might have enhanced soil nutrients through the weathering of basalt stones. Weathering would have made additional nutrients such as phosphorus and potassium available over the long term. Our documentation of carbon flecking within garden soil profiles (Stevenson *et al.* 2006) suggests that organic materials were added to increase fertility. This enrichment process may have been complemented by the incorporation of other nutrients to retard the depletion of soils. The weathering of basalt outcrops is a geologically slow process but the fragmentation of stone to generate lithic mulch, and the pulverization of that stone with durable basalt hoes, would have increased the surface area exposed to leaching by many orders of magnitude. This greater surface area to volume ratio accelerates the mineral weathering time and might have added minerals to the soil in just a few years. We will conduct a series of soil chemical assessments to confirm and quantify this process.

We will examine the Rapa Nui landscape on an island-wide scale to see how the type, frequency, and placement of gardens varied over time and space. Rapa Nui is highly variable with respect to soil types, rainfall amounts, wind velocities, and temperature. These variables depend upon the volcanic substrate, elevation and orientation of a particular locality. It is our operative premise that ancient farmers would have been intimately familiar with these variables and selected the rock garden form and crop type that would minimize overall investment while enhancing the probabilities that the crop would reach full maturity.

We have developed a GIS model for agricultural potential on Rapa Nui and in this project we will evaluate the model parameters and predictions with climate monitoring, field survey, and geomorphological excavations. In addition, we propose to chronologically date the use of the Rapa Nui landscape with radiocarbon and obsidian hydration dating and to identify crops grown in various garden settings through the analysis of starch grains found in soil and on stone tools (Torrence and Barton 2006; Horrocks and Wozniak 2007). These data will enable us to empirically document variable dryland agricultural development throughout the island.

Development of the GIS Model

We have developed a GIS model for the dryland agricultural potential of different areas of Rapa Nui that takes into account the variables of annual rainfall, age of the geological substrate, wind velocities, elevation, and air temperature. The parameters used to develop the model are based upon the best available estimates that for some variables (e.g., rainfall, which varies from an estimated annual value of 1200 mm at the coast to 2450 mm at the top of Terevaka) are poorly substantiated with regional quantitative data. Nevertheless, this preliminary model identifies areas of the island that could have sustained intensive dryland agriculture. The agricultural potential of a particular area is a function of the age of the geologic substrate and the amount of annual rainfall.

Research in the volcanic Hawaiian Islands (Vitousek et al. 2004) has determined that essential agricultural nutrients (such as phosphorus) are leached by rainfall. The older the substrate, the less rainfall can be sustained before a critical depletion of nutrients via leaching occurs. Volcanic substrates older than 300k are leached by rainfalls above the minimum annual rainfall on the island, which is 1200 mm. Substrates younger than 300k can sustain annual rainfalls up to 1400 mm, substrates younger than 230k can sustain 1500 mm, substrates younger than 200k can sustain 1600 mm, substrates younger than 175k can sustain 1800 mm, substrates younger than 40k can sustain 2000 mm, and substrates younger than 10k can sustain 2200mm. These parameters of rainfall and substrate age can be used to define six zones. Zones coded as 0 are either too old or receive too much rainfall to be agriculturally productive. The other coded zones could sustain dryland agriculture.

Evaluation of the Global Information System (GIS) Model

Over a three-year period, systematic field survey, geomorphological investigations, and environmental monitoring will be used to record settlement patterns, natural topography and features, and prehistoric gardens. Environmental sensors will also be installed to measure wind speed, rainfall, air temperature, ground temperature, and soil moisture. The chronology of landscape use will be monitored through the dating of excavations conducted at settlements and within agricultural fields. All of these data will be used to evaluate and refine the GIS model of dynamic agricultural development.

Archaeological Survey and Excavation

We propose to survey a total of sixteen 500-square-meter quadrants that will sample each of the distinct landscape forms on Rapa Nui. These survey units have been selected because they have not been impacted by current farming and rock removal. Each quadrant will be surveyed by close

interval (10-meter) pedestrian survey in teams of four persons. Differential GPS will be used to mark prehistoric surface archaeological features, garden boundaries, and any modern features. This will be followed by an assessment of the number and types of prehistoric gardens and sites. Up to six one-square-meter test units will be placed at domestic habitation sites or within gardens in each quadrant. Excavation units will be dry sieved through 0.5-inch hardware cloth and large volume soil samples will be collected for flotation. The profile of each test unit will undergo a geomorphological analysis to determine soil type, permeability and structure (see below). Soil samples will be selected for nutrient analysis and starch grain identification. Obsidian flakes will be retained for functional analysis, starch grain analysis, and obsidian hydration dating.

Environmental Monitoring: Soil Moisture, Soil Temperature, Wind Velocity

Rock gardens on Rapa Nui were developed to protect plants from high winds and rapid moisture loss. It has also been proposed that the rock ground cover, or lithic mulch, promoted water penetration, reduced temperature amplitude, and maintained elevated levels of ground moisture. All of these effects are highly probable but none have been quantitatively demonstrated over an annual cycle. A quantitative assessment of climate and soil properties over time is necessary to test our assumptions about the beneficial effects of rock gardens and to determine which parameters (e.g. temperature, moisture) were the most affected.

There are numerous types of rock gardens on Rapa Nui including boulder gardens, lithic mulched fields, surface rock pavements as well as gardens without rock cover that occur on different landforms at varied elevations (10-500 meters. As a result, the environmental relationships are complex. However, we can effectively model the variation in soil moisture, soil temperature, rainfall, and wind speed as a function of elevation and landform (e.g. hilltop, swale, open field) through the monitoring of conditions for a two-year period at different locations. Although the parameters will not provide representative long-term values, the relative values will allow us to compare regions of the island on a quantitative basis.

Computerized stand-alone weather stations (e.g. HOBO stations) that record ground temperature, soil water content, wind speed and direction, air temperature and relative humidity on a 1.5-hour basis will be placed at six different locations within our landscape categories. Stations will be situated within rock gardens and ground sensors positioned within and outside of gardens. A comparison of the annual trends in soil moisture and temperature will permit an evaluation of which landforms and garden types were the most effective at providing better growing conditions for crops. The HOBO sensors will be installed during the first month of the project and data will be remotely collected by radio at an office base station established on the island. The sensors will be monitored daily by S. Haoa to ensure that they are functioning properly.

Evaluation of Soils: Soil Nutrient Analyses

Recent research on Hawai'i by a team of scientists (including members of this project) has focused on the relationship between agricultural development and nutrient levels in the soils (Vitousek *et al.* 2004; Ladefoged and Graves 2005; Kirch *et al.* 2006; Hartshorn *et al*. 2006). This research convincingly shows that a complex relationship exists between the age of the lava substrate, the amount of rainfall in the area, and the degree of agricultural use of the region. Understanding this relationship on Rapa Nui will allow us to understand how the prehistoric islanders selectively used the landscape to increase agricultural production and reduce the risk of crop failure.

The volcanic island of Rapa Nui was formed by repeated eruptions ranging in age from 3,000,000 years ago to 10,000 years ago and is a complex mosaic of lava flows of different ages. In general, the older the geological substrate, the more nutrients in the surface soil will be depleted by rainfall over the millennia. The soils on more recent lava flows are likely to be nutrient rich but in areas of high rainfall, leaching may be accelerated and make the region less suitable for agriculture. Therefore,

our task is to determine which locations on Rapa Nui fall within the optimum range of nutrient and rainfall conditions for the production of crops such as sweet potato, taro, and yam.

The ranking of agricultural zones can be achieved through a program of soil chemical analysis. Multiple soil samples from distinct geological flows and different elevations will be tested for a range of nutrients, with an emphasis placed on the total available phosphorus (P) and other soil nutrients (i.e. Na, Ca, K, C, Sr). Our pilot study of this problem was conducted in 2005 and consisted of paired samplings of soils within and outside of ancient gardens on different geological substrates. Analysis of P levels indicates that Rapa Nui soils are generally impoverished on an island-wide level with concentrations generally below the 25 µg/g level determined to be close to the minimum for sweet potato production (Vitousek *et al.* 2004). However, areas of younger substrates generally have higher nutrient levels, and this may have influenced the use of these in the prehistoric landscape. Gardens in these areas generally have lower nutrient levels than the immediately surrounding non-garden areas. Other gardens have similar P levels to the immediately surrounding areas, and these gardens are generally found on moderate aged substrates at moderate rainfall levels. In contrast, other gardens have higher nutrient levels than the immediately surrounding areas, and these are generally found on older substrates at higher rainfall levels. These preliminary results suggest that farmers were in some cases depleting gardens of nutrients and in other cases enhancing nutrient levels. It will be important in this study to carefully determine the spatial and temporal attributes of both depleted and enriched gardens.

Our environmental investigation will focus on the processes of soil nutrient dynamics and physical erosion. We will apply a suite of soil and geochemical approaches in Rapa Nui that were developed to investigate intensive agriculture in Hawai'i (Chadwick *et al.* 1999, 2003; Vitousek *et al.* 2004; Hartshorn *et al.* 2006; Meyer *et al.* 2007). On Rapa Nui, as in Hawai'i, the rock chemistry is reasonably consistent in space and time, and well characterized – so we can calculate the dynamics of rock-derived elements in agricultural and non-agricultural areas with reference to an immobile index of elements. Erosion pathways will be characterized through field observations, and integrated using tracers of surface stability (Be isotopes, lead accumulation). Erosion and nutrient depletion (from harvest and leaching) and nutrient enhancement (from rock mulching in gardens) leave very different element and isotope profiles, and we will evaluate the strength of each independently from sample locations throughout the island.

We will excavate deep soil pits within and outside areas that supported intensive agricultural systems, and collect samples by horizon from the surface to bedrock. We propose to sample ~50 pits in this manner, distributed across the different age volcanoes, their land forms and the land use histories of Rapa Nui. These samples will be complemented by a larger number (~200) of integrated surface samples to 30 centimeters in depth that will be collected across the island, once we understand the deeper and more detailed profiles. This sampling approach will allow us to test ideas about the interacting roles of soil age and natural fertility, human land-use choices, as well as the degradation of soil properties (physical and chemical).

Soil analyses will focus on present soil fertility through measurement of resin-extractable P and for total C and N at Stanford University, and pH and exchangeable cations and cation exchange capacity (CEC) at the University of California, Santa Barbara (UCSB). We will also determine the long-term dynamics of soil fertility by comparing nutrient elements in intensively farmed soils to those in areas that were not farmed. We will then compare these to the concentration of nutrients in the lava that was the original source for the soils. The net loss or gain of nutrient elements from the soils will be determined by measurement with x-ray fluorescence spectrometry and calculated with reference to concentrations of an immobile index element such as Nb and Sr. We will assess levels of nutrient enhancement via rock garden mulching by evaluating Sr levels within and outside of gardens. If lithic mulching in gardens enhances nutrient levels, there should be increased levels of

Sr associated with the disintegration of locally available basalt rocks incorporated into the sediment. In contrast, in non-garden control areas, the Sr concentrations should be associated with air borne deposition from marine sources.

Finally, our soil analyses will tackle the question of physical erosion as opposed to nutrient depletion. Physical erosion can be identified in three ways: 1) at the landscape scale erosion leaves scars such as gullies that are identifiable and indicative of major local losses of soil, 2) at the soil profile scale, truncation of soil horizons can be identified and traced laterally across landscape elements, and 3) subtle erosion of surface layers can be quantified using isotopic tracers such as 210Pb and 137Cs (for modern rates) and meteoric 10Be (for integrated rates) that can be measured at UCSB. Isotopic tracers accumulate in soil surface horizons and the approach we will apply compares the quantity of isotope in reference soil profiles (non-eroded as determined by approaches 1 and 2) and suspect soil profiles.

Garden Function and Chronology

A chronology of landscape use will be provided by the AMS dating of carbon particles found within garden soils and by obsidian hydration dating (OHD) of artifacts removed from the Ap-horizon. We propose to process up to six AMS dates from garden and domestic site contexts within each survey unit (n=96). Realizing that this is not a statistically valid number of samples, we will process up to 100 obsidian hydration dates from each excavated context within each survey area (n=1600) by infrared measurement of the hydration layer (Stevenson *et al.* 2001). While OHD is problematic in other parts of the world, it is very useful in Rapa Nui where the geology is relatively simple and lithic imports are minimal. Recent research has resulted in significant advances in the method. A major achievement has been the development of high precision (e.g. SIMS calibrated) techniques of hydration rim measurement using infrared spectroscopy (Stevenson *et al.* 2001; Stevenson *et al.* 2004) where a very low absorbance measurement error results in a hydration layer thickness error of +/-0.03 um, allowing us to determine dates to within +/- 25 years. The mechanism of water diffusion in rhyolite glasses is well understood and is a function of glass structural water content (Zhang and Behrens 2000). This critical parameter can also be precisely determined by infrared spectroscopy (Stevenson et al. 2004) and used as a predictor variable of the rate of hydration for individual artifacts. To achieve accurate results care must be taken to use only subsurface specimens exposed to high levels (e.g. 98%) of relative humidity and to avoid thermal contexts that alter the hydration process (Anovitz *et al.* 2006). Soil temperature is also a critical variable, and has already been monitored with thermal cells on Rapa Nui. However, we will refine the modeling of ground temperature with the data from the HOBO stations. With these data, we will be able to calculate an effective hydration temperature for different depths within the soil for distinct depositional contexts (e.g. the different categories of lithic gardens). Applied to newly excavated specimens, these temperatures should greatly increase the accuracy and precision of chronological determinations. In addition, we will be able to evaluate previous OHD estimates made using salt based thermal cells in the early 1990s.

We will also assess the types of cultigens grown in the different categories of gardens through the analysis of starches from garden contexts (Torrence 2006). Exploratory studies (Horrocks and Wozniak 2007) have shown that starches from taro, yam and sweet potato are well preserved within soils on Rapa Nui. The identification and relative frequency of starch grains will provide a comprehensive view of what was grown in the garden throughout its use. Starch grains removed from the surface of obsidian flakes, routinely encountered within the Ap-horizons, can provide a temporal look and what was cultivated once the obsidian artifact has been dated. Such an approach may allow the identification of regional planting patterns and changes in the crops grown over time at a particular location. We will analyze a total of 100 samples to look for patterns of plant cultivation.

RESULTS AND OPPORTUNITIES

This research project is focused on Rapa Nui, yet the issues we address are global. Rapa Nui is often depicted as a model for global environmental degradation, and the research will investigate the ways in which agriculture was adapted to and transformed a pristine island environment. Rapa Nui is an ideal study area for documenting the dynamic nature of these processes. It is a relatively small isolated island where the volcanic biogeochemical matrix can be examined with precision and accuracy. The short period of occupation (ca. 800-1,000 years) and relatively low levels of outside interaction facilitate an understanding of socio-political transformations. Our approaches and models will provide insights into the relationships between a dynamic ecosystem, agricultural development, and sociopolitical evolution. These findings will enable social scientists to understand the impact of people both on the local island level, and the wider global scale.

The results of our project will be shared with the local population as well as with the international community. While on the island each year, we present a lecture to the local organization of tourist guides that incorporates the results of the previous field season. These data are used to update their tours and thereby improve the service of the tourist industry, the mainstay of the island economy. In addition, a lecture is given at the local high school to upper level students. Our reports are submitted to the local government officials and we also relate to them in oral fashion what we have discovered about the archaeological record. The National Park and the thousands of sites it contains are under great pressure from agricultural expansion and we believe that our new information may increase awareness about the uniqueness of archaeological resources and help in land management decisions.

The results of this research are published in several formats. Each year the results of the previous field excavation are summarized in the form of a field report and distributed to the Chilean governmental agencies (National Library of Chile, Father Englert Museum). A summary of the field results is also published in the *Rapa Nui Journal*, a semi-annual publication with a wide distribution to both the public and professionals involved in Easter Island research. Research results have also been published in professional journals (*Antiquity, Journal of New World Archaeology, Oceania, Clava*) and scholarly books (*Maunga Tari: An Upland Agricultural Complex, The Evolution of Island Societies, Rapanui Studies*) and will continue to be released in this format.

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EXPEDITION PACKING CHECKLIST

Essential Items

- ☐ **This Expedition Briefing**
- ☐ **Photocopies of your passport, flight itinerary and credit cards in case the originals are lost or stolen; the copies should be packed separately from the original documents**
- ☐ **Passport and/or visa (if necessary)**
- ☐ **Certification of inoculation (if necessary)**

Required Items

Clothing/Footwear for Fieldwork

- ☐ Lightweight, quick drying, long-sleeved shirts
- ☐ Durable work pants (e.g. jeans)
- ☐ Well worn-in hiking boots or work boots
- ☐ Brimmed hat to protect your face from the sun
- ☐ Poncho or rain jacket

Excavation Kit

Note: Each participant will be required to furnish his/her own excavation kit containing small items of personal equipment.

- ☐ Small daypack/rucksack to hold excavation supplies
- ☐ 4-5-inch flat pointed masonry trowel with sharpened (single bevel) edges (Marshalltown brand in the US is best and available online at www.marshalltown.com/artrwbuy.html and in hardware stores) - trowels do not come sharpened
- ☐ 3-meter tape
- ☐ Waterproof marker
- ☐ 2 pencils
- ☐ 2 pens
- ☐ 25 paper bags (lunch bag type)
- ☐ 25 plastic sealable freezer bags (e.g. Ziploc or similar brand)
- ☐ Strong, durable work gloves
- ☐ 1 complete newspaper
- ☐ Inexpensive whisk broom and metal dust pan (Team I only)
- ☐ Ball of string (Team II only)

Other Field Supplies

- ☐ Drybag or additional plastic sealable bags to protect personal equipment from dust, water, etc.
- ☐ Sunscreen lotion with SPF 30 or higher
- ☐ Insect repellent spray for mosquitoes
- ☐ Canteen or water bottle

Clothing/Footwear for Leisure

- ☐ T-shirts
- ☐ Shorts
- ☐ Comfortable shoes
- ☐ Light sweater/sweatshirt for evenings
- ☐ Swimsuit
- ☐ Set of clothing to keep clean for church, recreational time, end of expedition, etc.

Personal Supplies

Note: The hotel will supply bathroom towels and bedding.

- ☐ Personal toiletries (biodegradable soaps and shampoos are encouraged)
- ☐ Partial roll of toilet paper for the field
- ☐ Antibacterial wipes or lotion (good for "washing" hands while in the field)
- ☐ Personal First Aid kit (e.g. anti-diarrhea pills, antibiotics, antiseptic, itch-relief, pain reliever, bandages, moleskin, etc.) and any personal medications you may need

Miscellaneous

- ☐ Spending money (see *Other Advice/Information* in Section 9 *'Before You Leave'* for advice)
- ☐ Camera, film/digital memory storage, extra camera battery
- ☐ Luxury food/drink items to share with the group (e.g. beef jerky, granola, chocolate, wine, scotch, etc. – see Section 8 *'Food'* for more details)

Optional Items

- ☐ Seat cushion or kneeling pad for excavation work (recommended)
- ☐ Beach towel
- ☐ Pocket knife (pack in checked luggage, not carry-on)
- ☐ Flashlight or headlamp with extra batteries and extra bulb
- ☐ Earplugs for light sleepers
- ☐ Some duct tape (always handy, though a whole roll is not necessary)